

Lecture #2: Build LLM with PyTorch.

#1: All primitives to train a model

#2: bottom-up ~~from~~ [tensors to model to optimizer]

#3: efficiency case of ~~the~~ resources

memory (GB)
compute (FLOPS)

↓

(15 TB)
Num of Tokens

(70B) Model size $\Rightarrow$ compute $\Rightarrow$ (1000 H100s)

example:

total flops: $6 * 70e9 * 15e12 \rightarrow$ num of tokens
↓
num of param.

H100_flop_per_sec = $1979e12/2$.

↓
mfu = 0.5

↓
Flops_per_day = H100_flop_per_sec * mfu * 1024 * 60 * 60 * 24

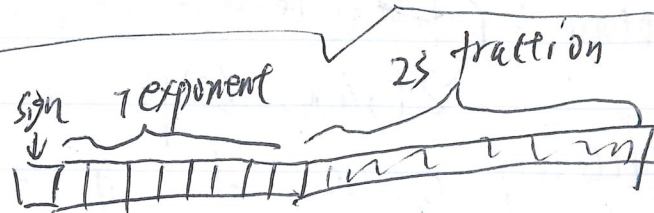
↓
days = total_flops / flops_per_day

Q1: how many hours to train a model?

Q2: what's the largest model that you can train on ...?

32-bit float / tensor. basic: ∇ / tensor. memory
 weights / parameters / gradients / activations / optimizer
 of state
 model state /

[matrix]

32-bit float (by default): 
 { 16-bit float (half precision)

BF 16 float: Brain Floating = dynamic range

same memory as FP16 with as FP32

FP-8: E4M3 and E5M2

$\Rightarrow$ mix-precision training $\left\{ \begin{array}{l} \text{FP32: inefficient} \\ \text{FP16/BF16:} \end{array} \right. \left\{ \begin{array}{l} \text{unscalable} \\ \text{overflow} \\ \text{underflow} \end{array} \right.$

e.g. $\text{FNN} = \left\{ \begin{array}{l} \text{backward path} \Rightarrow \text{FP8 / FP16} \\ \text{forward path} \end{array} \right.$

Transformer { Attention: calculations $\Rightarrow$ FP32.

Computing accounting

CPU $\Leftrightarrow$ GPU $\left[\begin{array}{l} \boxed{\text{SMU}} \quad \boxed{\text{SMU}} \\ \boxed{\text{SMU}} \quad \text{streaming multiprocessing} \end{array} \right.$